

Agentic Healthcare Assistant

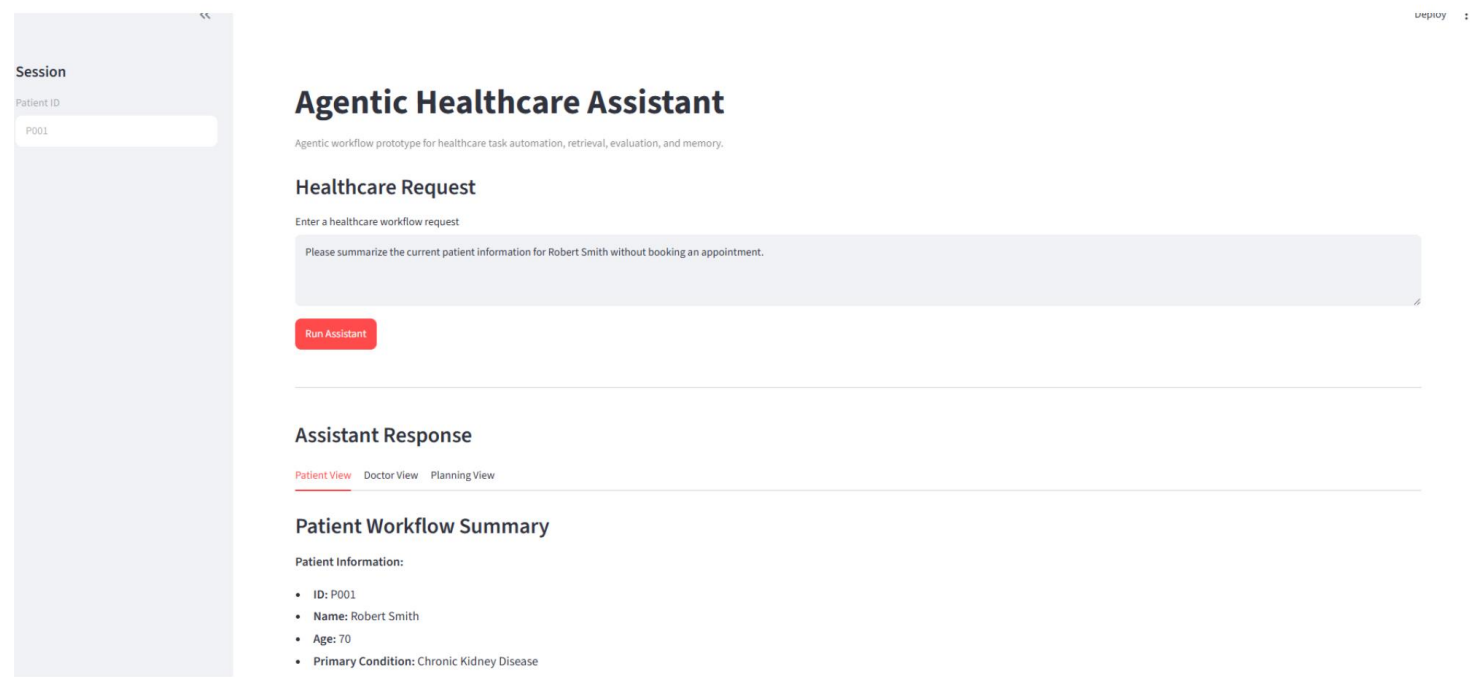
A patient-scoped workflow prototype for retrieval, evaluation, memory, and safe task automation.

What this deck demonstrates

A real application run, the evidence behind the run, and the controls that keep the workflow bounded.

The patient view makes the workflow understandable at a glance

The system starts with a plain-language request and returns a bounded, patient-scoped response.



Evidence: patient ID, request, response tabs, and workflow summary are visible in one view.

The response is grounded in a patient record before synthesis

The record view separates source facts from the model-generated summary.

Medical Record:

- Record ID: R001
- Diagnoses: Chronic Kidney Disease
- Medications: Lisinopril 10 mg daily
- Treatment Notes: Monitor kidney function and blood pressure
- Relevant Alerts: Demo alert: Medication allergy information requires verification
- History Notes:
 1. Step 2.5 restart persistence verified on 2026-09-04
 2. Phase 4 Step 4.2 add and update persistence verified on 2026-09-05

Appointment Information:

- Status: None scheduled

Selected Doctor:

- Status: None selected

Relevant Alerts

Demo alert: Medication allergy information requires verification.

Grounding controls

- Patient-scoped retrieval
- Medical record fields remain inspectable
- Alerts and history are visible
- No appointment is implied unless requested

Design implication: the reviewer can trace the summary back to structured inputs.

The deck shows how quality is measured—not just how the UI looks

Metrics are paired with formulas, sources, and evaluation outcomes.

Operational Performance

Aggregated from persisted TOOL_EXECUTION events. Skipped booking steps are excluded from the attempt denominator.

Booking Success Rate	Successful Bookings	Booking Attempts	Failed Bookings
81.82%	9	11	2

Formula: $\text{successful book_appointment executions} / \text{book_appointment attempts}$

Source: `persisted_tool_execution_events`

Run records scanned: 53 | Malformed records skipped: 0

Response Precision

Claim-level precision calculated from persisted semantic-evaluation records. Authority, safety, and relevance flags are excluded.

Response Precision	Supported Claims	Verifiable Claims
98.86%	87	88

Formula: $87 / 88 = 98.86\%$

Evaluation Basis: Persisted semantic-evaluation records. Structured and retrieved response claims are counted; authority, safety, and relevance flags are not counted as response claims.

Evaluation Status

Rubric-readable model-response and operational tool metrics for the latest workflow run.

Model Response Metrics

Deterministic Evaluation	Semantic Evaluation	Accepted for Memory
PASS	PASS	YES

Tool Success Metrics

Tools Attempted	Tools Successful	Tools Failed	Tool Success Rate
4	4	0	100%

Appointment Summary

No appointment was created in this workflow.

Operational success and response precision

The workflow makes tool ordering and skipped actions explicit

A summary-only request retrieves and validates information while safely skipping appointment actions.

Memory

This response passed evaluation and was stored in patient memory.

Patient ID: P001

Last Specialty: Nephrology

Last Appointment ID: A001

Stored Turns: 3

Workflow Diagnostics

Tool Execution Events

Ordered per-tool success/failure events persisted with the latest pipeline run.

Sequence	Tool	LangGraph Node	Capability	Outcome	Execution Evidence
1	get_patient_by_id	retrieve_patient	Patient DB lookup	SUCCESS	workflow_state.patient populated
2	get_medical_record_by_patient	retrieve_medical_record	EHR / Patient DB lookup	SUCCESS	workflow_state.medical_record populated
3	search_patient_summaries	retrieve_patient_summary	Patient-scoped FAISS summary retrieval	SUCCESS	workflow_state.patient_summaries populated
4	search_medlineplus / search_knowledge fallback	retrieve_knowledge	Trusted external retrieval with FAISS fallback	SUCCESS	workflow_state.retrieved_documents populated
5	find_doctors_by_specialty	find_doctors	Doctor Schedule API adapter lookup	SKIPPED	Skipped because appointment_requested is false
6	book_appointment	book_appointment	Doctor Schedule API adapter booking (deterministic demo)	SKIPPED	Skipped because appointment_requested is false

Evidence: four retrieval tools succeed; doctor search and booking are skipped because appointment_requested is false.

External knowledge is shown with source and retrieval context

The system identifies trusted sources and records when they were retrieved.

Retrieved Sources

1. Chronic Kidney Disease

Source Name: MedlinePlus.gov — National Library of Medicine

Source Type: Trusted external health-topic web service

Source URL: <https://medlineplus.gov/chronickidneydisease.html>

Retrieved At (UTC): 2026-09-07T23:28:49.211520+00:00

Retrieved Content:

You have two kidneys , each about the size of your fist. Their main job is to filter your blood. They remove wastes and extra water, which become urine. They also keep the body's chemicals balanced, help control blood pressure, and make hormones. Chronic kidney disease (CKD) means that your kidneys are damaged and can't filter blood as they should. This damage can cause wastes to build up in your body. It can also cause other problems that can harm your health. Diabetes and high blood pressure are the most common causes of CKD. The kidney damage occurs slowly over many years. Many people don't have any symptoms until their kidney disease is very advanced. Blood and urine tests are the only way to know if you have kidney disease . Treatments cannot cure kidney disease , but they may slow kidney disease . They include medicines to lower blood pressure, control blood sugar, and lower cholesterol. CKD may still get worse over time. Sometimes it can lead to kidney failure. If your kidneys fail, you will need dialysis or a kidney transplantation. You can take steps to keep your kidneys healthier longer: Choose foods with less salt (sodium) Control your blood pressure; your health care provider can tell you what your blood pressure should be Keep your blood sugar in the target range, if you have diabetes Limit the amount of alcohol you drink Choose foods that are healthy for your heart: fruits, vegetables, whole grains, and low-fat dairy foods Lose weight if you are overweight Be physically active Don't smoke NIH: National Institute of Diabetes and Digestive and Kidney Diseases

2. Kidney Diseases

Source Name: MedlinePlus.gov — National Library of Medicine

Source Type: Trusted external health-topic web service

Source URL: <https://medlineplus.gov/kidneydiseases.html>

Retrieved At (UTC): 2026-09-07T23:28:49.211520+00:00

Retrieved Content:

You have two kidneys , each about the size of your fist. They are near the middle of your back, just below the rib cage. Inside each kidney there are about a million tiny structures called nephrons. They filter your blood. They remove wastes and extra water, which become urine. The urine flows through tubes called ureters. It goes to your bladder, which stores the urine until you go to the bathroom. Most kidney diseases attack the nephrons. This damage may leave kidneys unable to remove wastes. Causes can include genetic problems, injuries, or medicines. You have a higher risk of kidney disease if you have diabetes, high blood pressure, or a close family member with kidney disease . Chronic kidney disease damages the nephrons slowly over several years. Other kidney problems include: Cancer Cysts Stones Infections Your doctor can do blood and urine tests to check if you have kidney disease . If your kidneys fail, you will need dialysis or a kidney transplant. NIH: National Institute of Diabetes and Digestive and Kidney Diseases

Memory is patient-scoped, evaluated, and auditable

The workflow records why information was read, written, or skipped without exposing conversation content.

Memory Diagnostics

Memory Access Events

Patient-scoped READ, WRITE, and SKIP metadata. Request and response content are never included in this trace.

Timestamp (UTC)	Action	Patient ID	Outcome	Source	Validation Basis	Reason	Stored Turns	Content Exposed
2026-09-06T18:19:21.007354Z	SKIP	P001	SKIPPED	PatientMemoryStore	deterministic_and_semantic_acceptance	evaluation_gate_rejected	None	☐
2026-09-06T18:24:11.760143Z	SKIP	P002	SKIPPED	PatientMemoryStore	deterministic_and_semantic_acceptance	patient_memory_not_found	None	☐
2026-09-06T20:03:01.500796Z	SKIP	P001	SKIPPED	PatientMemoryStore	deterministic_and_semantic_acceptance	evaluation_gate_rejected	None	☐
2026-09-06T20:14:00.199191Z	READ	P001	SUCCESS	PatientMemoryStore	deterministic_and_semantic_acceptance	validated_turn_supplied_to_follow_up_prompt	1	☐
2026-09-06T20:38:33.508176Z	WRITE	P001	SUCCESS	PatientMemoryStore	deterministic_and_semantic_acceptance	evaluation_accepted_and_persisted	2	☐
2026-09-06T22:18:04.989329Z	SKIP	P001	SKIPPED	PatientMemoryStore	deterministic_and_semantic_acceptance	evaluation_gate_rejected	None	☐
2026-09-07T20:08:39.803654Z	SKIP	P001	SKIPPED	PatientMemoryStore	deterministic_and_semantic_acceptance	evaluation_gate_rejected	None	☐
2026-09-07T20:22:29.318721Z	SKIP	P001	SKIPPED	PatientMemoryStore	deterministic_and_semantic_acceptance	evaluation_gate_rejected	None	☐
2026-09-07T21:09:24.163231Z	SKIP	P001	SKIPPED	PatientMemoryStore	deterministic_and_semantic_acceptance	evaluation_gate_rejected	None	☐
2026-09-07T21:10:35.909771Z	SKIP	P001	SKIPPED	PatientMemoryStore	deterministic_and_semantic_acceptance	evaluation_gate_rejected	None	☐

No workflow errors recorded.

> Deterministic Evaluation Details

> Semantic Evaluation Details

> Workflow State

> Interpreted Intent

Controls shown: patient scope, validation basis, evaluation gate, and no workflow errors.

Specialty matching stays downstream of patient grounding

The workflow uses the requested specialty and patient context before exposing doctor or appointment actions.

Memory

This response passed evaluation and was stored in patient memory.

Patient ID: P001

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Doctor View evidence frame

- Patient context is already populated
- Specialty match is a distinct workflow step
- Doctor lookup is conditional
- Booking remains a separate, explicit action

Evidence note: this slide combines the captured execution trace with the verified doctor/booking boundary.

The plan is descriptive; execution remains gated

REQ-016 is satisfied by returning ordered plan_steps without authorizing tool execution.

1

Retrieve patient information

2

Validate patient identity

3

Match specialty and retrieve knowledge

4

Book only when explicitly requested

5

Synthesize and present the response

Planning boundary

The plan describes the sequence. The graph decides which tools execute.

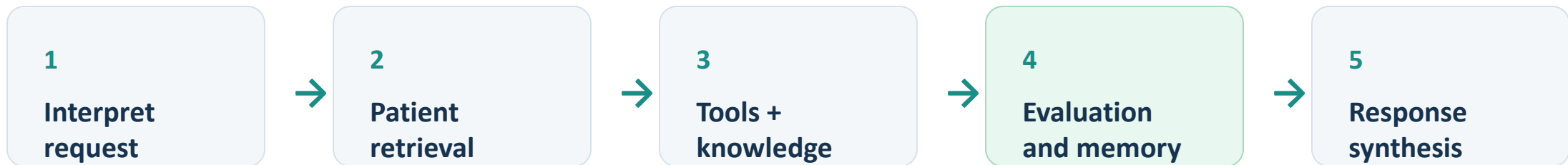
Evidence reconstruction based on the verified RequestIntent plan_steps and workflow diagnostics.

The system is organized as a bounded evidence pipeline

Each stage has a clear responsibility and produces inspectable state for the next stage.

Architecture principle

Reliability is supported by explicit boundaries: structured contracts, conditional tool execution, evaluation gates, provenance, and patient-scoped memory.



This is an explanatory architecture view; implementation evidence remains in the subsequent application captures.

Major requirements map directly to visible evidence

The reviewer can move from a requirement to a concrete screen, metric, or diagnostic.

CAPABILITY	VISIBLE EVIDENCE
Patient grounding	Patient View + Medical Record
Planning	Planning View + ordered plan_steps
Doctor matching	Doctor View + specialty boundary
Safe execution	Workflow Diagnostics + skipped booking
Evaluation	Evaluation Status + precision metrics
Memory governance	Memory Diagnostics + patient scope

The compliance register remains the authoritative requirement record; this slide is the reviewer’s navigation aid.

The prototype is useful because its boundaries are visible

The submission distinguishes demonstrated capability from clinical authority.

Responsible-use boundaries

- Synthetic demonstration data
- Educational workflow prototype
- Not a medical diagnostic system
- External sources require verification
- Human oversight remains necessary

Submission readiness

- ✓ 40/40 requirements satisfied
- ✓ 125 regression tests passed
- ✓ Python compilation passed
- ✓ Secrets excluded from release
- ✓ README, register, and evidence included

What the reviewer should conclude

01 The workflow is visible

The patient view connects request, retrieval, planning, and response.

02 The evidence is traceable

Metrics, sources, memory events, and tool outcomes are inspectable.

03 The boundaries are explicit

Synthetic data, no-medical-advice limits, and skipped actions are clear.

Recommended submission set: final ZIP + this evidence PDF + compliance register.